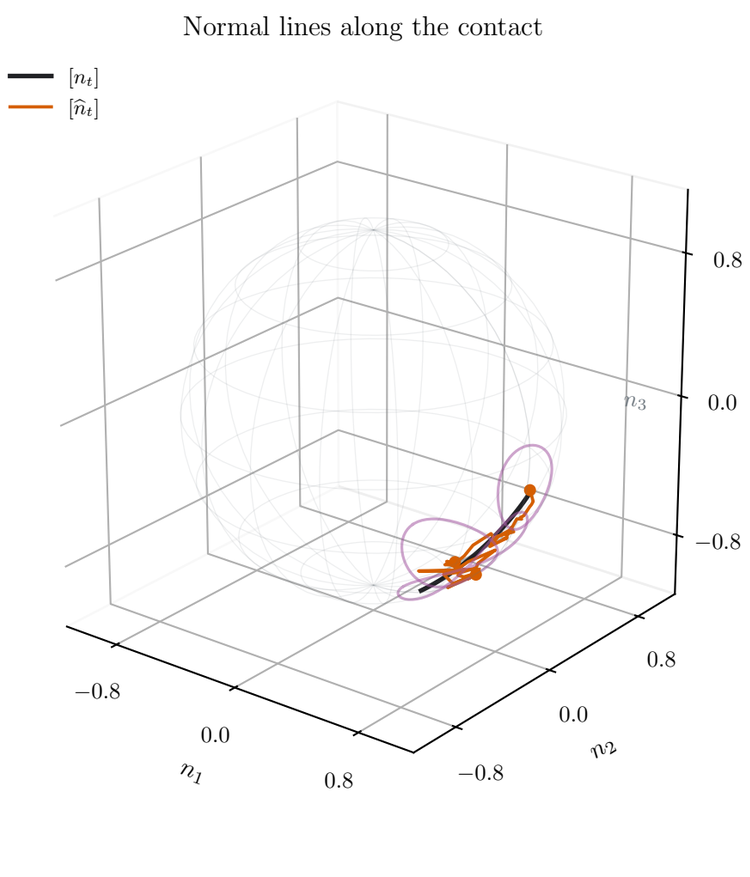
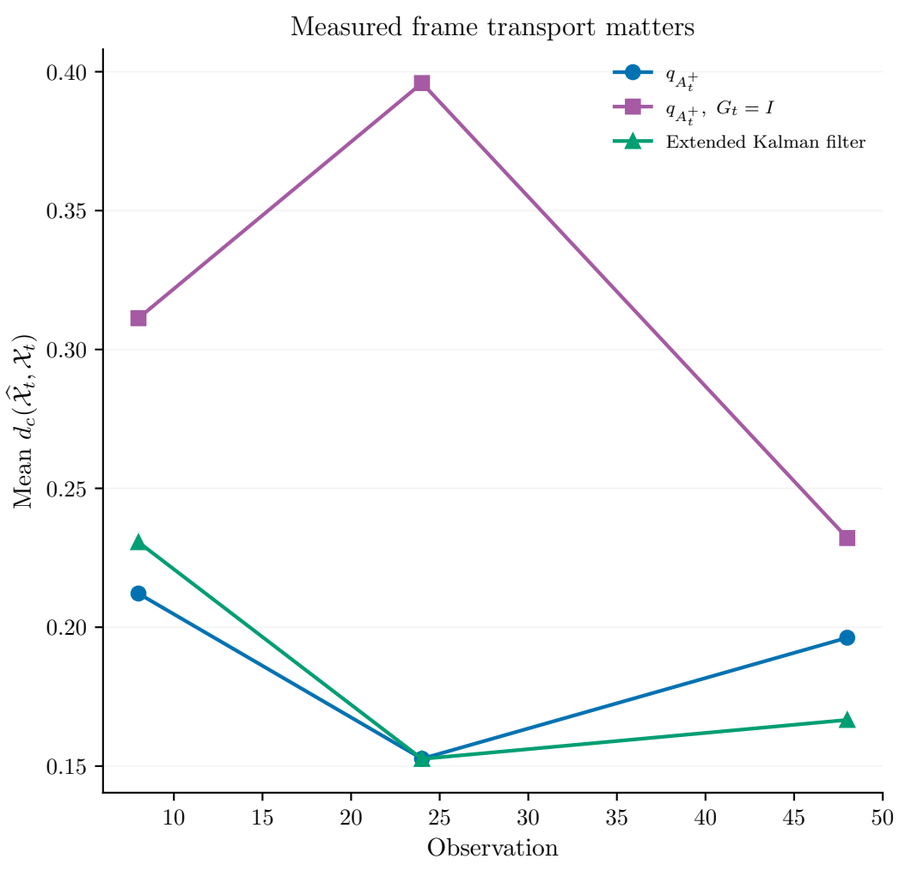
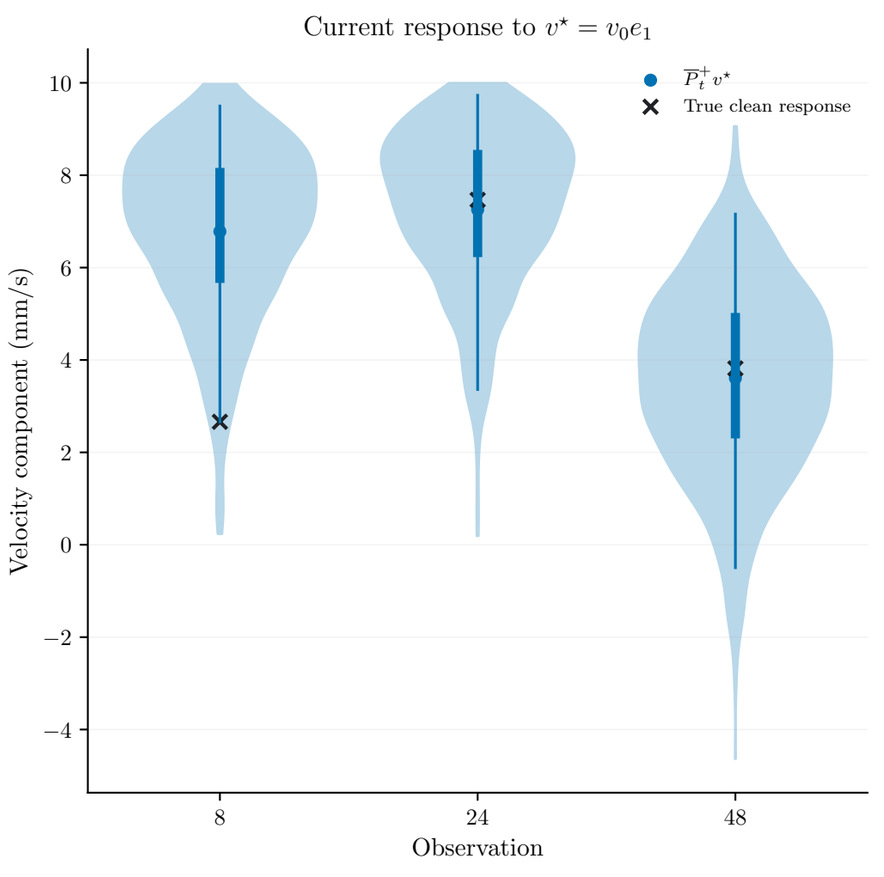

Normal lines along the contact
$[n_t]$
$[\widehat{n}_t]$
n_1
n_2
n_3
−0.8
0.0
0.8


Measured frame transport matters
$q_{A_t^+}$
$q_{A_t^+},\ G_t = I$
Extended Kalman filter
Mean $d_c(\widehat{\mathcal{X}}_t, \mathcal{X}_t)$
Observation
0.15
0.20
0.25
0.30
0.35
0.40
10
15
20
25
30
35
40
45
50


Current response to $v^\star = v_0 e_1$
$\overline{P}_t^+ v^\star$
True clean response
Velocity component (mm/s)
Observation
−4
−2
0
2
4
6
8
10
8
24
48